

Quasi-Poisson models

Last time: quasi-Poisson models

Poisson:

- + Mean = λ_i
- + Variance = λ_i

quasi-Poisson:

- + Mean = λ_i
- + Variance = $\phi\lambda_i$
- + Variance is a linear function of the mean

Last time: quasi-Poisson models

Poisson:

- + $\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim N(0, 1)$
- + Use $N(0, 1)$ for hypothesis testing and confidence intervals for one parameter
- + Comparing nested models (drop-in-deviance): χ^2 test

quasi-Poisson:

- + $SE(\hat{\beta}_j)_{QP} = \sqrt{\hat{\phi}} \times SE(\hat{\beta}_j)_{Pois}$
- + $\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)_{QP}} \sim t_{n-p}$
- + Use t_{n-p} for hypothesis testing and confidence intervals for one parameter
- + Comparing nested models: F test

Class activity

Work on the class activity (handout).

Class activity

Which model do you think is more appropriate for the data?

Class activity

Is there a difference in the average number of cigarettes smoked between smokers with some college education and smokers with no college education, after accounting for sex, age, diabetes, and BMI?

Model 1:

	Estimate	Std. Error	z value	Pr(> z)
...				
education2	0.0164626	0.0126312	1.303	0.192462
...				

Model 2:

(Dispersion parameter for quasipoisson family taken to be 5.519388)

Nested tests with quasi-Poisson models

```
m2 <- glm(cigsPerDay ~ male + age + education + diabetes + BMI,  
          data = smokers, family = quasipoisson)  
summary(m2)
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.837192	0.115099	24.650	< 2e-16	***
male	0.456170	0.026095	17.481	< 2e-16	***
age	-0.006806	0.001595	-4.267	2.07e-05	***
education2	0.016463	0.029675	0.555	0.579	
education3	0.016427	0.037706	0.436	0.663	
education4	-0.015033	0.040477	-0.371	0.710	
diabetes	-0.025398	0.092606	-0.274	0.784	
BMI	0.005001	0.003304	1.513	0.130	

Research question: Is there a relationship between education level and the number of cigarettes smoked per day, after accounting for sex, age, diabetes, and BMI?

Nested tests with quasi-Poisson models

Research question: Is there a relationship between education level and the number of cigarettes smoked per day, after accounting for sex, age, diabetes, and BMI?

- + In Poisson regression, we would use a likelihood ratio test
- + However, the quasi-Poisson model includes the estimated dispersion, $\hat{\phi}$. We need to use an **F-test** instead

Nested tests with quasi-Poisson models

```
m1 <- glm(cigsPerDay ~ male + age + education +  
          diabetes + BMI,  
          data = smokers, family = quasipoisson)  
m2 <- glm(cigsPerDay ~ male + age + diabetes + BMI,  
          data = smokers, family = quasipoisson)
```

```
m1$deviance
```

```
## [1] 11539.56
```

```
m2$deviance
```

```
## [1] 11543.84
```

```
m1$df.residual
```

```
## [1] 2004
```

```
m2$df.residual
```

```
## [1] 2007
```

Nested tests with quasi-Poisson models

```
m1$deviance
```

```
## [1] 11539.56
```

```
m2$deviance
```

```
## [1] 11543.84
```

```
pf(0.258, 3, 2004, lower.tail=F)
```

```
## [1] 0.8556648
```

Limitations of quasi-Poisson models

Important: quasi-Poisson models do *not* describe a full distribution for the response (there is no "quasi-Poisson" distribution). They describe *only* the mean and variance of the response.

As a result, one cannot calculate probabilities for a quasi-Poisson:

```
m2 <- glm(cigsPerDay ~ male + age + diabetes + BMI,  
          data = smokers, family = quasipoisson)  
rootogram(m2)
```

Error in prodist.glm(object) :
family is not associated with a full probability distribution

Class activity

Work on the class activity (handout).